#HW 12

Graded

Student

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Total Points

94 / 100 pts

Question 1

(no title) **30** / 30 pts

- ✔ **− 0 pts** Correct

Question 2

(no title) **9** / 10 pts

- **− 0 pts** Correct
- **− 1 pt** minimum theoreotical power input value?
- **− 1 pt** wrong Tc and Th temp. taken
- ✔ **− 1 pt** is 3 hp sufficient or not?

Question 3

(no title) **15** / 15 pts

- ✔ **− 0 pts** Correct
- **− 3 pts** reversible heat pump calculation?
- **− 5 pts** incomplete solution

Question 4

(no title) **13** / 15 pts

- **− 0 pts** Correct
- ✔ **− 2 pts** Pressure at end of isothermal expansion = 76.1 kPa
- **− 4 pts** pressure not calculated

Question 5

(no title) **15** / 15 pts

- ✔ **− 0 pts** Correct
- **− 2 pts** Case (a) is irreversible

Question 6

(no title) **12** / 15 pts

- **− 0 pts** Correct
- ✔ **− 3 pts** incomplete solution

Questions assigned to the following page: 3, 2, 4, and 1

2) $COP = \frac{\dot{Q}_{in}}{\dot{W}_c} = \frac{30\,000\ Btu/h}{(3\cancel{hp})(2545\ \frac{Btu/h}{\cancel{hp}})} = 3.93$

$COP_{ideal} = \frac{T_c}{T_h - T_c} = \frac{(70+458.67)\cancel{R}}{(90+458.67)\cancel{R} - (70+458.67)\cancel{R}}$

$= 24$

$\dot{W}_c = \frac{\dot{Q}_{in}}{COP_{ideal}} = \frac{30\,000\ Btu/h}{24} = 1250\ Btu/h$

$= 1250\ \cancel{Btu/h} \cdot \frac{1h}{2545\ \cancel{Btu/h}} = 0.49\ hp$

3) $\dot{Q}_{in} = 740\ Btu/min \cdot \frac{1055.06\ J}{Btu} \cdot \frac{1}{60s} = 13.048\ kW$

a) $P_{elec} = \dot{Q}_{in} = 13.048\ kW$

b) $P_{elec} = \frac{\dot{Q}_{in}}{COP} = \frac{13.048\ kW}{3} = 4.349\ kW$

c) $COP_{carnot} = \frac{(68+458.67)R}{(68+458.67)\cancel{R} - (32+458.67)\cancel{R}}$

$= 14.7$

$P_{elec} = \frac{\dot{Q}_{in}}{COP_{carnot}} = \frac{13.048\ kW}{14.7} = 0.889\ kW$

4) $COP_{carnot} = \frac{T_c}{T_H - T_c} = \frac{300\cancel{K}}{(600-300)\cancel{K}} = 1$

$W_{net} = Q_H - Q_C$

$\frac{Q_H}{Q_C} = \frac{T_H}{T_C} \Rightarrow Q_C = \frac{Q_H T_C}{T_H} = \frac{(250\ kJ/kg)(300K)}{600K}$

$Q_C = 125\ kJ/kg$

Questions assigned to the following page: 4 and 5

$W_{net} = Q_H - Q_C = (250 - 125)\,kJ/kg = 125\,kJ/kg$

$\frac{P_1}{V_1} = \frac{T_H}{T_C}$

$\frac{V_3}{V_1} = \frac{T_H}{T_C}$

$P_3 = P_1 \cdot \frac{V_1}{V_3} \Rightarrow P_3 = P_1 \cdot \frac{T_C}{T_H} = (325\,kPa)\frac{(300K)}{(600K)}$

$P_3 = 162.5\,kPa$

5)a) $\dot{W}_{net} = \dot{Q}_H - \dot{Q}_C$

$= 600\,kW - 400\,kW$

$= 200\,kW$

$\eta = \frac{\dot{W}_{net}}{\dot{Q}_H} = \frac{200\,kW}{600\,kW} = 0.33 \cdot 100\% = 33\%$

$\eta_{carnot} = 1 - \frac{T_C}{T_H} = 1 - \frac{400K}{1200K} = 0.67 \cdot 100\% = 67\%$

$\eta < \eta_{carnot}$ } irreversibly

b) $\dot{W}_{net} = 600\,kW - 0\,kW = 600\,kW$

$\eta = \frac{600\,kW}{600\,kW} = 1 \cdot 100\% = 100\%$

$\eta_{carnot} = 67\%$

$\eta > \eta_{carnot}$ } impossible

c) $\dot{W}_{net} = 600\,kW - 200\,kW = 400\,kW$

$\eta = \frac{400\,kW}{600\,kW} = 0.67 \cdot 100\% = 67\%$

$\eta_{carnot} = 67\%$

$\eta = \eta_{carot}$ } reversibly

Question assigned to the following page: 6

QH (kJ)	TH (K)	QC (kJ)	TC (K)	σ cycle (kJ/K)	η (%)
750	1500	100	500	-3	0.866667